

FULL STACK DEVELOPMENT

DATE:20/5/26

DAY:3

TOPICS COVERED:

COMPILER AND INTERPRETER

FEATURES OF JAVA

JVM(JAVA VIRTUAL MACHINE)

JRE(JAVA RUN TIME ENVIRONMENT)

JDK(JAVA DEVELOPMENT KIT)

INTEGRATED DEVELOPMENT KIT

COMPILER AND INTERPRETER:

COMPILER: Compiler compiles a whole program at a time and it converts it into machine language. and it creates a classfile for that program.

INTERPRETER: Interpreter means it reads a code line by line and executes a code. It shows syntax error while typing a code. JVM is also a part of interpreter. Bytecode executes a code line by line and allows a portability in write once run anywhere.

FEATURES OF JAVA:

Object oriented, platform independent, simple and easy to learn, secure, robust, multithreaded, high performance, distributed, portable.

JVM(JAVA VIRTUAL MACHINE):

The JVM is the engine to run the program and it takes the bytecode and executes the program and it's a platform independent on windows, linux, mac.

What JVM does:

- Loads code: takes compiled bytecode (.class files)
- Verify code: ensures bytecode is safe
- Executes code: convert byte code into machines
- Manages memory: run garbage collector to free unused memory

JRE(JAVA RUNTIME ENVIRONMENT):

It creates an environment to run a java program. it includes the jvm and libraries(pre-built code).

What's inside:

Jvm executes the java

Core libraries

Supporting files

Eg:

Jre jvm+libraries

Need to run java programs, but not write/compiler them

JDK(JAVA DEVELOPMENT KIT):

It develops a java application and it includes a tools, write, compile, debug and run

What's inside jdk:

Jre

Jvm+core libraries

Development tools:

Compiler – converts source code(.java) into bytecode(.class)

Debugger-finds and fix errors

Documentation-generate docs from code

What jdk does:

Write code.(in editor/ide)

Compile code(java)

Run code(using JRE and JVM)

Without jdk, you can run java programs(with jre), but you cannot create/compile them.

WORKFLOW

Jvm-runs java byte code

Jre-jvm+libraries(to run java apps)

Jdk-jre+tools(to develop java apps)

INTEGRATED DEVELOPMENT EDITOR:

An ide is a software application that provides a complete environment to write,edit,compile,debug and run code easily.

Its basically a programmers workspace.

Main features:

- Code editor
- Compile/interpreter
- Debugger
- Project management
- Others tools

Integrated development tools:

- Eclipse
- Intellij idea
- Net beans
- VS code

How jvm uses together:

Interpreter:excutes bytecode immediately

Jit:speeds up frequent code by converting it to native code

Jvm uses both:interpreter for startup+jit for performance

Native machine code:0s and 1s cpu understable language

